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Throughput of ASCON Compared with Popular IoT Encryption Algorithms

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Cover Page Footnote

Lightweight, quantum resilient ciphers are increasingly needed for Internet of Things (IoT) devices that cannot afford the time or energy cost of AES. This study benchmarks the three NIST-standard ASCON variants, ASCON-128, ASCON 128a, and ASCON 80pq, against the AES modes found in current protocols (AES-128-CTR/GCM and AES-256-CTR/GCM). Encryption and decryption times were measured over 1,000 iterations on an Arduino Uno R4 for a 256 byte payload and associated data (AD) sizes 0–256 B. ASCON 128/128a ran 4.0–6.0 times faster than AES 128 CTR, while ASCON 80pq outperformed AES 256 CTR by 6.1 times (all $t(999) < -28\,000$, $p < 2 \times 10^{-308}$). Across the AD range, ASCON throughput scaled linearly yet stayed below 800 μs , whereas AES GCM remained flat at 3,555 and 3,931 μs ; projections show ASCON keeps its speed advantage until AD exceeds $\approx 2\text{--}3\text{ kB}$, well above typical IoT metadata. Replacing AES + HMAC stacks in LoRaWAN, CAN, or tactical mesh radios with ASCON AEAD can cut code, latency, and energy, while ASCON 80pq delivers post quantum security without the overhead of 256 bit AES. All code and data are openly archived for replication.

Throughput of ASCON Compared with Popular IoT Encryption Algorithms

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Abstract

Lightweight, quantum-resilient ciphers are increasingly needed for Internet-of-Things (IoT) devices that cannot afford the time or energy cost of AES. This study benchmarks the three NIST-standard ASCON variants, ASCON-128, ASCON-128a, and ASCON-80pq, against the AES modes found in current protocols (AES-128-CTR/GCM and AES-256-CTR/GCM). Encryption and decryption times were measured over 1,000 iterations on an Arduino Uno R4 for a 256-byte payload and associated-data (AD) sizes 0–256 B. ASCON-128/128a ran 4.0–6.0 times faster than AES-128-CTR, while ASCON-80pq outperformed AES-256-CTR by 6.1 times (all $t(999) < -28\,000$, $p < 2 \times 10^{-308}$).

Across the AD range, ASCON throughput scaled linearly yet stayed below $800\,\mu\text{s}$, whereas AES-GCM remained flat at $3,555$ and $3,931\,\mu\text{s}$; projections show ASCON keeps its speed advantage until AD exceeds $\approx 2\text{--}3\,\text{kB}$, well above typical IoT metadata. Replacing AES + HMAC stacks in LoRaWAN, CAN, or tactical mesh radios with ASCON-AEAD can cut code, latency, and energy, while ASCON-80pq delivers post-quantum security without the overhead of 256-bit AES. All code and data are openly archived for replication.

Keywords: ASCON, lightweight cryptography, IoT security, benchmarking, post-quantum algorithms

In a prior study conducted under the VICEROY Scholars program, the authors demonstrated that the ASCON encryption algorithm significantly outperformed AES-128-CTR, the Advanced Encryption Standard (AES), in both encryption and decryption speeds, establishing it as a strong candidate for use in constrained Internet of Things (IoT) and embedded system environments. However, as the cryptographic landscape evolves, new considerations, including the rise of quantum computing and the increasing importance of authenticated encryption, prompt a broader analysis. The present paper extends that benchmark to multiple ASCON variants and AES modes, adds authenticated-encryption scenarios, and situates the results in a post-quantum context. Because AES was not designed for modern IoT constraints or post-quantum threats, ASCON, recently selected by the National Institute of Standards and Technology (NIST) for lightweight cryptography, is a promising alternative.

This study: (1) benchmarks the encryption and decryption runtime and throughput for ASCON-128/128a/80pq versus AES-128-CTR/GCM and AES-256-CTR/GCM, (2) examines how each algorithm scales with associated-data size, (3) test whether observed differences are statistically significant, and (4) assessed ASCON's viability in applications with power restrictions, limited computational resources, or those that demand post-quantum resistance.

Background

Threats in the IoT Space

It is widely recognized that encryption flaws and configuration errors are recurring concerns in both traditional and IoT-focused systems. Many active IoT protocols rely on ciphers like AES in counter mode (CTR), which, although cryptographically sound, are susceptible to side-channel attacks and more power-intensive than alternatives. Differential power analysis recovered a full AES-128 key in just 100 traces (O'Flynn & Chen, 2014). Fault-injection attacks deliberately introduce voltage or clock glitches that force transient hardware errors during encryption. Using this technique, researchers have retrieved a full 128-bit AES key with only two pairs of correct and faulty ciphertexts on an embedded target (Piret & Quisquater, 2003; Kim & Quisquater, 2008).

ASCON

History

ASCON, developed in 2014 by Graz University of Technology, was explicitly designed to meet the needs of IoT, embedded systems, and other low-power applications. Initially submitted to the CAESAR Competition, it was recognized as the best option for lightweight cryptographic use cases (Bernstein, 2019). In 2018, NIST called for cryptographic algorithms optimized for resource-constrained devices, prompting ASCON to be submitted alongside 56 other candidates. After rigorous evaluation, ASCON was announced as the winner in early 2023. By late 2024, the first draft of the ASCON standard was released, solidifying its role as a leading solution for lightweight cryptography (Turan et al., 2024).

Design

ASCON is a lightweight cryptographic solution specifically designed for devices with limited resources, such as those in IoT environments. At its core, it is a family of authenticated encryption algorithms with associated data (AEAD) that encrypts the payload and verifies the

integrity of unencrypted associated data. As a cryptographically secure symmetric algorithm, it offers the same level of protection as AES for the same key size.

All ASCON variants share a 320-bit sponge state and a bit-sliced substitution-permutation network. After one 12-round initialization that absorbs the key, nonce, and IV, the cipher runs in duplex mode: plaintext blocks are XORed into the “rate” half of the state, permuted for 6 or 8 rounds, then read out as ciphertext; a 12-round finalization injects the key again and outputs a 128-bit tag. Variant parameters adjust rate and security margin: ASCON-128 (64-bit rate, six reduced rounds) is the baseline; ASCON-128a doubles the rate to 128 bits and compensates with eight reduced rounds; ASCON-80pq extends the key to 160 bits for post-quantum margin while keeping the original round schedule (Dobraunig et al., 2021).

Security Properties

ASCON’s bit-sliced 5-bit S-boxes and word-level XOR/rotate operations prevent data-dependent memory access, enhancing resistance to timing and power-analysis attacks. The secret key is accessed only during the 12-round initialization and finalization; once diffused into the 320-bit state, intermediate state leakage does not reveal the key (Dobraunig et al., 2021), although unmasked implementations may leak in about 500 traces (Gross et al., 2017). Follow-up studies have extracted keys from microcontroller implementations of ASCON with 8,000 traces (Weissbart & Picek, 2023). First-order masked ASCON adds only 8,000 gate equivalents, about a 3.1 times increase over the unprotected version, yet shows no detectable leakage after 10 million traces in multiple tests (Kaps & Gaj, 2022). This suggests ASCON effectively withstands side-channel attacks. Fault-injections face similar limitations: Statistical Ineffective Fault Analysis recovered just two key bits from 280 faulty authentication tags (Ramezanzpour et al., 2020), indicating strong practical robustness under standard countermeasures.

Methodology and Research Design

Algorithm Selection

The study benchmarks two pairs: ASCON-128/128a vs AES-128-CTR to reflect today’s IoT stacks that default to 128-bit AES in counter mode (LoRa Alliance, 2017; Silicon Labs, 2018, 2023), and ASCON-80pq vs AES-256-CTR to model a post-quantum setting in which Grover’s algorithm halves effective key strength (Sarah & Peter, 2024). AES-256 offers margin by brute force, while ASCON-80pq delivers a similar margin (Dobraunig et al., 2021). Comparing each ASCON variant to its AES counterpart isolates the cost and speed impact of switching ciphers under identical security targets.

Hardware Selection

The performance testing was run on an Arduino Uno R4 WiFi ABX00087 (Renesas RA4M1, 32 kB RAM). The Arduino R4 was selected due to its prevalence in IoT and “maker” spaces. It was also ideal to simulate the real-world, resource-constrained environment

for which ASCON was intended.

Collection Procedure

The Arduino connects to a computer via the COM port. The comlog.py script utilizes a Python serial listener to communicate with the Arduino, sending the start code "START". The Arduino then outputs the algorithm name and executes a 1000-iteration loop of encryption and decryption using a random key, plaintext, and nonce (if applicable), timing each operation with the `micros()` function. This randomization prevents biases or artificial optimizations that arise from reusing the same inputs, ensuring that the performance metrics reflect real-world usage. After each operation, it calculates the duration and returns the results over serial. Finally, the Arduino sends a stop signal "#" to indicate the completion of operations.

Data Analysis

After all iterations, the raw logs of the serial communication are cleaned and separated into individual CSV files. We computed the mean time per 256-byte block, per-byte time, and throughput (in bytes per second), along with their standard deviations and 95% confidence intervals. One-tailed independent *t*-tests evaluated whether ASCON variants were significantly faster than their AES counterparts ($\alpha = .01$); full *t*-tables appear in Appendix B. The Python library SciPy was used to evaluate all p-values. Due to floating-point precision limits, some p-values were evaluated to be 0.0. To represent the probability, the smallest number that Python can distinguish from 0 was substituted, 2.2×10^{-308} . The complete code can be found linked in Appendix A.

Results

Table 1 shows that the ASCON variants run between 319 μ s and 500 μ s, whereas the AES variants all exceed 1,900 μ s

Table 1

Mean, Standard Deviation, and Throughput for Each Encryption and Decryption Operation

Algorithm	Mean Enc Time (μ s/256B)	Enc. Std. Dev. (μ s)	Enc Throughput (B/s)	Mean Dec Time (μ s/256B)	Std. Dev. Dec. (μ s)	Dec Throughput (B/s)
AES-128-CTR	1,910	0.9	134,030	1,906	0.9	134,250
AES-128-GCM	3,555	1.5	72,012	3558	1.5	71,947
AES-256-	2,668	1.4	95,948	2665	1.4	96,060

CTR						
AES-256-GCM	3,930	0.8	65,130	3933	0.8	65,078
ASCON- 128	422	1.4	606,664	470	1.4	543,927
ASCON- 128a	319	1.3	802,068	371	1.3	689,934
ASCON-80pq	440	1.4	581,693	493	1.4	518,826

Figure 1 graphically represents this data. Error bars are added to each bar chart to show the 95% confidence interval.

Figure 1.

Bar chart showcasing the encryption and decryption times of cryptographic algorithms.

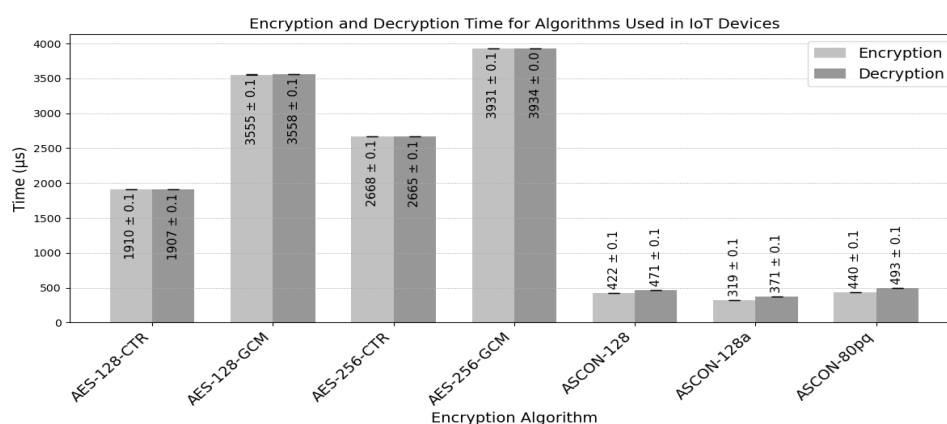


Table 2 shows that at 256 B of associated data, the ASCON lines top out at 744, 543, and 761 μs, while both AES-GCM lines remain flat at roughly 3,555 and 3,931 μs.

Table 2

Mean and Standard Deviation for Associated Data Supporting Algorithms with Variable Associated Data Sizes Encryption Times

Associated Data Size (Bytes)	AES-128-GCM		AES-256-GCM		ASCON- 128		ASCON-128a		ASCON-80pq	
	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ
0	3555	1.5	3931	0.8	422	1.4	319	1.3	440	1.4

1	3623	1.4	3999	0.5	435	1.4	335	1.4	453	1.4
2	3623	1.4	3999	0.5	435	1.4	335	1.3	453	1.4
4	3623	1.4	3999	0.5	436	1.4	336	1.3	454	1.4
8	3623	1.4	3999	0.5	442	1.4	334	1.3	460	1.4
16	3555	1.4	3931	0.8	452	1.4	345	1.3	470	1.4
32	3555	1.5	3931	0.9	471	1.4	359	1.4	489	1.4
64	3555	1.5	3931	0.8	511	1.4	385	1.4	529	1.4
128	3555	1.4	3931	0.8	587	1.4	436	1.4	604	1.3
256	3555	1.4	3931	0.8	744	1.2	543	1.4	761	1.2

As visualized in **Figure 2**, the three ASCON curves rise linearly ($R^2 > .99$) but stay under 800 μs ; the AES-GCM curves form a flat band around 3,555 and 3,931 μs .

Figure 2

Encryption time in microseconds of a 256-byte payload plotted against the size of associated data in bytes.

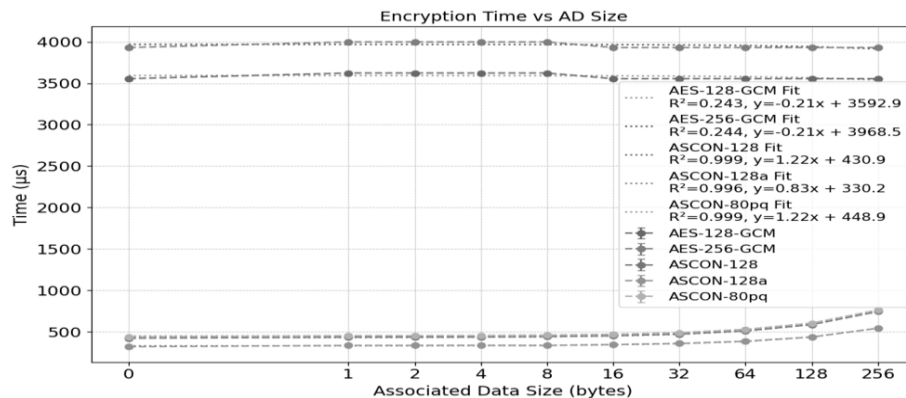


Table 3 shows that decryption mirrors encryption: all ASCON variants remain $\leq 817 \mu\text{s}$ at 256 B, whereas AES-GCM variants hold steady near 3,558 and 3,934 μs .

Table 3

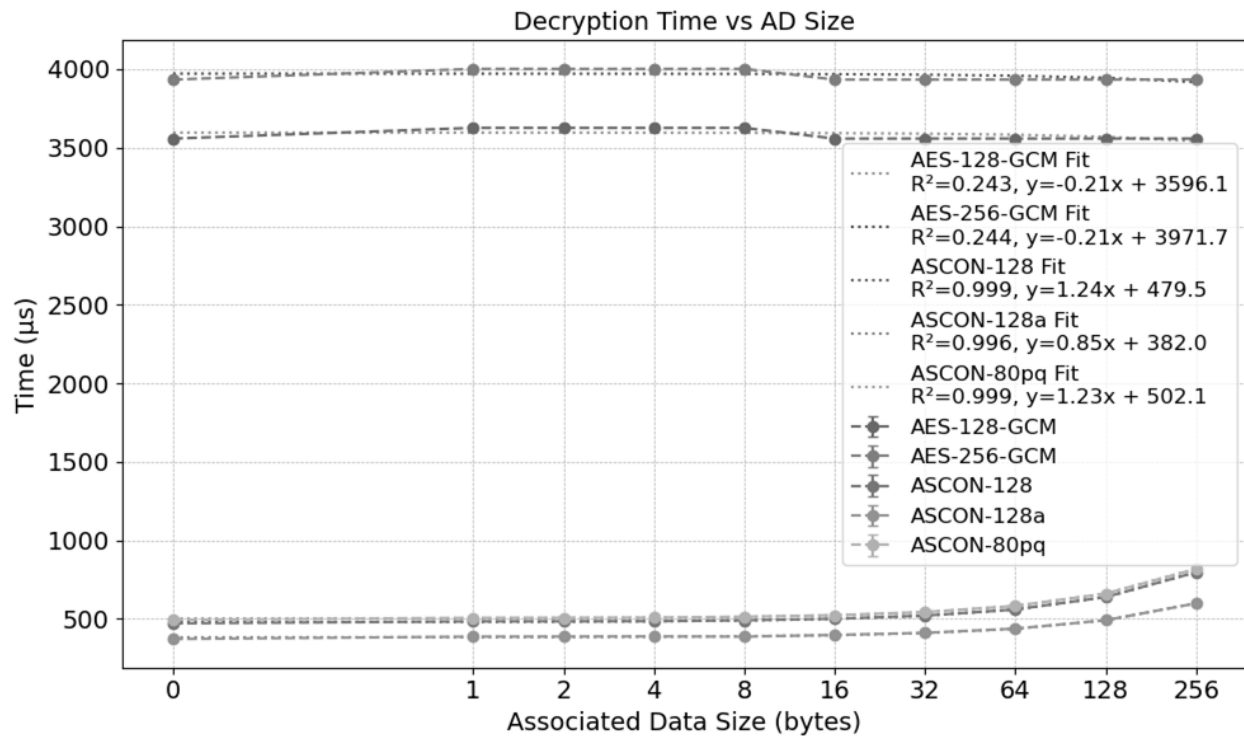
Mean and Standard Deviation for Associated Data Supporting Algorithms with Variable Associated Data Sizes Decryption Times

Associated Data Size (Bytes)	AES-128-GCM		AES-256-GCM		ASCON-128		ASCON-128a		ASCON-80pq	
	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ
0	3558	1.5	3934	0.8	471	1.4	371	1.3	493	1.4
1	3626	1.4	4002	0.4	483	1.4	387	1.4	506	1.4
2	3626	1.4	4002	0.4	484	1.4	387	1.4	507	1.4
4	3626	1.4	4002	0.4	485	1.4	388	1.4	508	1.4
8	3626	1.4	4002	0.4	491	1.4	386	1.4	513	1.4
16	3558	1.4	3934	0.8	500	1.4	397	1.4	523	1.4
32	3558	1.5	3934	0.9	520	1.4	410	1.4	543	1.4
64	3558	1.5	3934	0.8	560	1.4	437	1.4	582	1.3
128	3558	1.5	3934	0.8	638	1.4	491	1.4	660	1.3
256	3558	1.5	3934	0.7	796	1.2	598	1.4	817	1.1

Figure 3 confirms the same linear-versus-flat pattern for decryption, keeping the ASCON lines well below the AES band across all AD sizes.

Figure 3

Encryption time in microseconds of a 256-byte payload plotted against the size of associated data in bytes.



Discussion and Analysis

Comparison in the IoT Space

ASCON-128 and ASCON-128a both beat AES-128-CTR decisively: one-tailed tests returned $t = -28,187/-27,536$ for ASCON-128 and $t = -31,469/-30,470$ for ASCON-128a ($p < 2 \times 10^{-308}$; Table B1). The average speedups were 4.5 times for encryption and 4.0 times for decryption for ASCON-128, and 6.0 times for encryption and 5.1 times for decryption for ASCON-128a. Those margins translate to lower energy draw and shorter latency on lightweight devices. Because 128a doubles the sponge rate and adds two extra permutation rounds, it delivers the largest real-time benefit while still meeting 128-bit security targets (Dobraunig et al., 2021).

Post-Quantum Comparison

Benchmarking the post-quantum pair showed that ASCON-80pq encrypts and decrypts substantially faster than AES-256-CTR: one-tailed tests gave $t = -36,730$ for encryption and $-35,281$ for decryption ($p < 2 \times 10^{-308}$; Table B2). Mean speed-ups were 6.1 times for encryption and 5.4 times for decryption. ASCON-80pq delivers comparable quantum-resilient security with far lower latency and power draw, making it the preferred alternative for constrained nodes that must remain secure as quantum capabilities mature (Dobraunig et al., 2021).

Comparison with Associated Data

Associated data (AD) lets AEAD ciphers authenticate header or routing bytes, typically < 120 B in common IoT protocols, without encrypting them. One-tailed tests confirmed that ASCON-128/128a encrypted and decrypted faster than AES-128-GCM ($t < -46\,000$, $p < 2.2 \times 10^{-308}$; Table B3) and that ASCON-80pq out-performed AES-256-GCM ($t < -66\,000$; Table B4). Runtime for ASCON-128, 128a, and 80pq rose linearly with AD size, yielding $R^2 > .99$ in both coefficient sets (Tables C1 and C2), consistent with their sponge duplexing step that re-permutes after every AD block. By contrast, AES-GCM remained essentially flat because its GHASH authenticator handles each 128-bit AD block in constant time while the AES-CTR core is AD-independent. Extrapolating the fitted lines shows that ASCON maintains its throughput edge until AD exceeds $\approx 2\text{--}3$ kB, far above any realistic IoT header, underscoring ASCON's efficiency advantage for resource-constrained deployments.

Applications for Current and Future Use

Improving LoRaWAN and Other IoT Networks

LoRaWAN's star-of-stars architecture uses three AES-protected keys, AppKey, NwksKey, and AppSKey, for join, network MAC, and payload encryption, respectively (LoRa Alliance, 2017; Trend Micro, n.d.). Swapping those AES functions for a single ASCON-AEAD call would combine encryption and authentication, cut code, and lower CPU cycles on battery-powered nodes. A field prototype that replaced AES with ASCON-128 in a theft-detection sensor saw a 42–45 % throughput drop (Amelia & Hartejo, 2025); our controlled benchmarks show the opposite, ASCON-128 runs 4–6 times faster than AES-CTR/GCM on a full 256-byte LoRa packet, suggesting the prototype's slowdown was an implementation issue, not a cipher limitation.

Securing the CAN Bus

The Controller Area Network (CAN) Bus carries tiny payloads (8 B Classic, 64 B CAN-FD) and has tight real-time deadlines, which makes software-based AES impractical (Nguyen & Chen, 2024). An Artix-7 FPGA prototype running ASCON-128 met CAN-FD timing with throughput roughly 100 times faster than a microcontroller build, while keeping power and logic area low (Nguyen & Chen, 2024). These results show that ASCON can add authenticated encryption to in-vehicle messages and, by extension, to other hard-real-time buses without violating latency or resource budgets.

Tactical Mesh Networks

Mobile ad-hoc radios such as Persistent Systems' WaveRelay and the goTenna Pro X create infrastructure-free meshes for voice, video, and situational-awareness data. Both protect traffic with a two-step stack, AES-256-CTR for secrecy plus HMAC-SHA-512 for integrity, adding latency and draining battery on vehicle, drone, and body-worn nodes (Persistent Systems, 2012). Replacing that pair with ASCON-80pq call would provide encryption and authentication in one pass and slash CPU cycles. Benchmarks in this study show ASCON-80pq running 5–6 times faster than AES-256-CTR on 32-kB microcontrollers. This margin would translate directly into longer field runtime and lower heat dissipation for mesh radios operating on limited power budgets.

Conclusion

This study benchmarked ASCON-128/128a/80pq against the AES configurations most common in IoT, AES-128-CTR/GCM and AES-256-CTR/GCM, on an Arduino Uno R4. Across every test-bed (baseline payloads, 0–256 B associated data, post-quantum pair), the ASCON variants encrypted and decrypted faster than their AES counterparts. ASCON-128a delivered the highest throughput, making it the best fit for real-time tasks, while ASCON-80pq combined post-quantum margin with a 6 times speed-up over AES-256-CTR. Although ASCON's runtime grows linearly with associated-data size, its crossover point with AES-GCM lies well above the metadata lengths typical in low-power networks. Hence, the speed advantage holds in practice.

These findings support replacing AES with ASCON in constrained devices where latency, energy, and future-proof security are important. With ASCON now a NIST-standardized algorithm for lightweight cryptography, developers have a credible, well-vetted alternative, optimized for modern IoT systems. Next steps include repeating the benchmarks on ARM Cortex-M, RISC-V, FPGA, and legacy AVR platforms, as well as adding power-draw and memory-footprint profiling. Our team is already integrating ASCON into a LoRa-based low-power wide-area mesh for search-and-rescue sensors, which will serve as a real-world proof of concept.

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